Global Lease Contract System

Databases & Webservices / CO-560-B

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Intended Functionality:

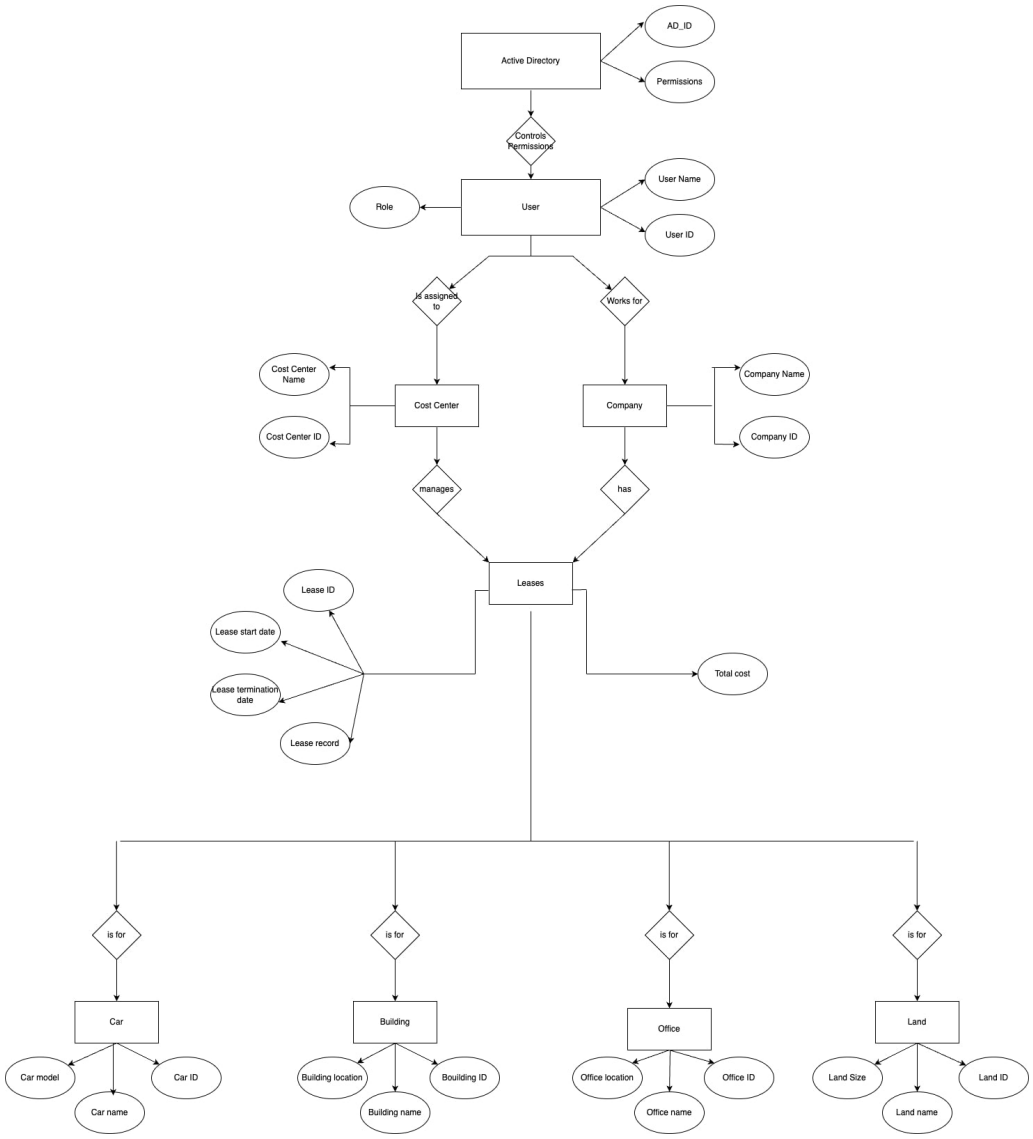
The Global Lease Contract System is designed to help companies manage and track their leased assets, such as cars, buildings, offices, and land. The system provides detailed information about each lease, including contract duration, associated costs, and billing charges. End-users, such as cost center managers and company representatives, can interact with the system to view and manage data related to their cost centers. They can make adjustments to lease data such as modifying termination dates, updating cost information, and adding new leased assets. Users can also request to join additional cost centers or companies to access more datasets, with the visibility of data controlled by a user principal name set up in an Active Directory list. The system ensures that only authorized users can view or modify data based on their roles and assigned permissions.

Entity-Relationship (ER) Diagram

The ER diagram for the Global Lease Contract System consists of multiple entities and relationships involving leases, companies, cost centers, users, and assets. Four ISA hierarchies (triangles) represent inheritance structures for different types of leased assets:

1. Car
2. Building
3. Office
4. Land

All these asset types inherit from a general *Lease* entity, which holds common attributes like lease start date, duration, and total cost. Users are associated with specific companies and cost centers, and their permissions to view and modify data are based on the information stored in the Active Directory.



<u>Entity Relation Diagram of Global Lease Contract System</u>

User Interactions

1. User Log-in: for users to register and log in to the DBMS.
2. User Dashboard: Users can view all leased assets and their associated contract details, such as contract duration, cost, and termination dates, based on their assigned permissions.
5. Modify lease information: Users can edit lease-related data, including termination dates, costs, and charges, for assets within their authorized scope.
3. Join additional cost centers/companies: Users can request access to additional companies or cost centers to view their lease data, subject to approval and validation.
4. Submit changes: Users can submit changes to lease data, with the system validating the legality of the input (e.g., termination dates cannot precede the current date).
5. Illegal actions, such as modifying data without proper authorization, will result in an error message.
6. Notifications: User gets reminded of relevant changes and chartings of their leasing contracts